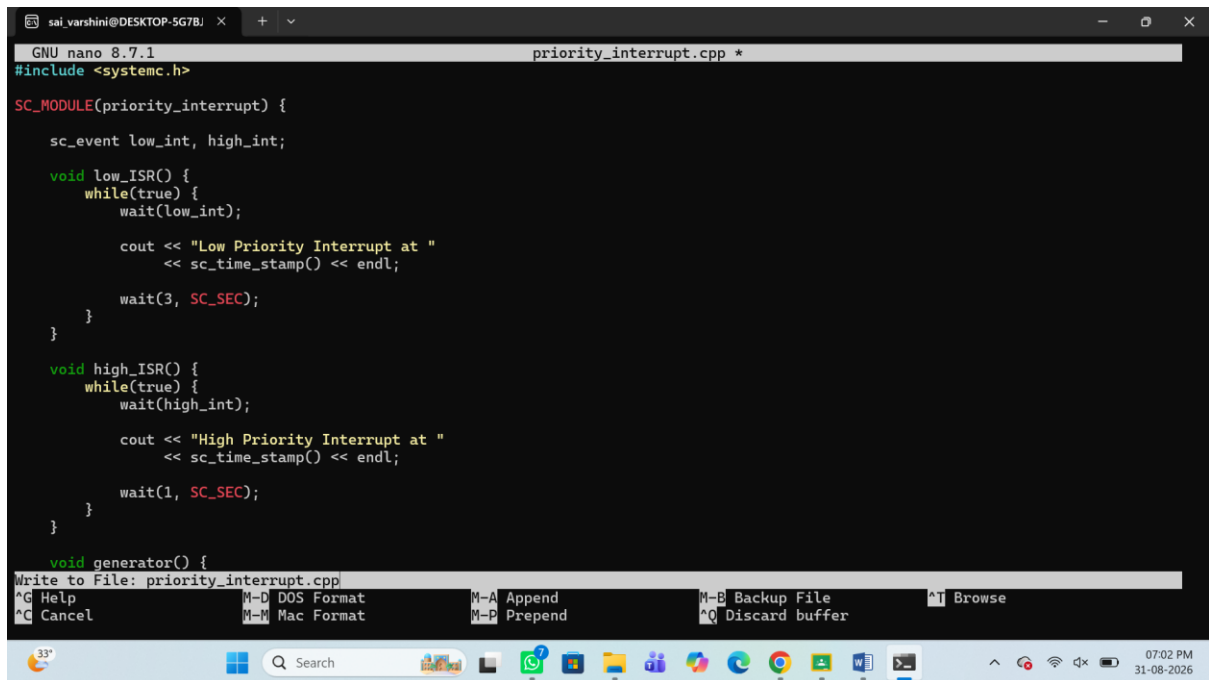


Exp No: 29

PRIORITY-BASED INTERRUPT HANDLING



```
GNU nano 8.7.1 priority_interrupt.cpp *
#include <systemc.h>

SC_MODULE(priority_interrupt) {

    sc_event low_int, high_int;

    void low_ISR() {
        while(true) {
            wait(low_int);

            cout << "Low Priority Interrupt at "
                 << sc_time_stamp() << endl;

            wait(3, SC_SEC);
        }
    }

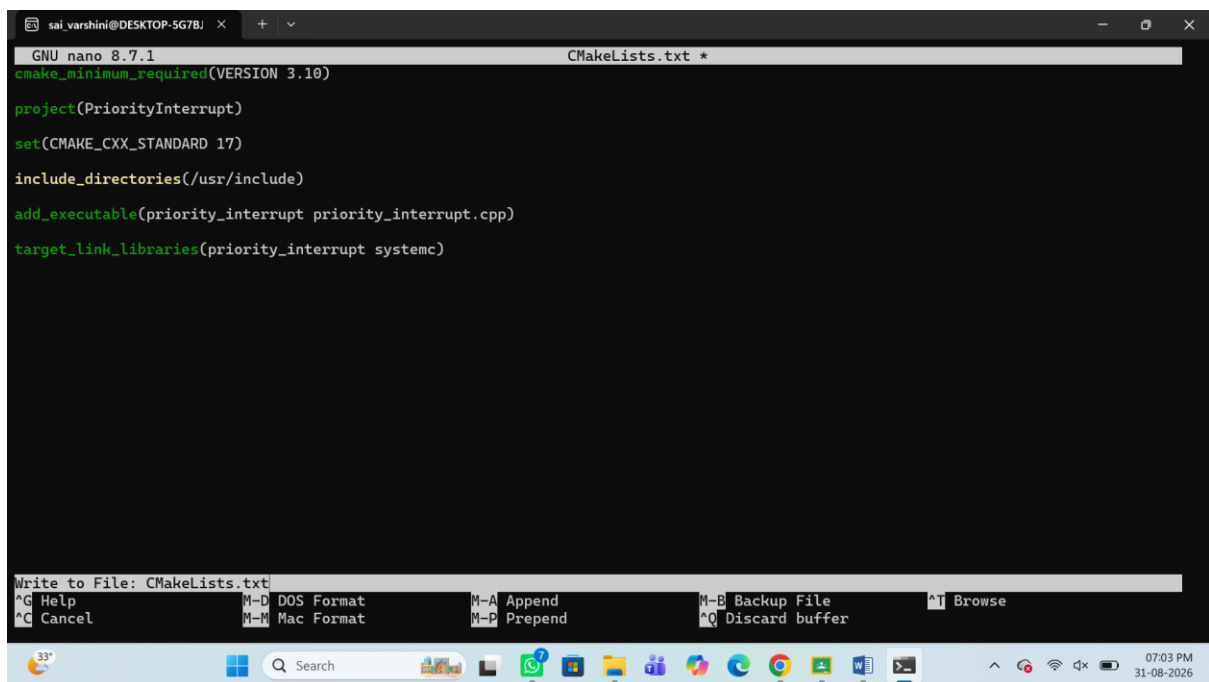
    void high_ISR() {
        while(true) {
            wait(high_int);

            cout << "High Priority Interrupt at "
                 << sc_time_stamp() << endl;

            wait(1, SC_SEC);
        }
    }

    void generator() {
        Write to File: priority_interrupt.cpp
        ^G Help      M-D DOS Format  M-A Append     M-B Backup File ^T Browse
        ^C Cancel    M-M Mac Format  M-P Prepend    M-Q Discard buffer

```



```
GNU nano 8.7.1 CMakeLists.txt *
cmake_minimum_required(VERSION 3.10)

project(PriorityInterrupt)

set(CMAKE_CXX_STANDARD 17)

include_directories(/usr/include)

add_executable(priority_interrupt priority_interrupt.cpp)

target_link_libraries(priority_interrupt systemc)

Write to File: CMakeLists.txt
^G Help      M-D DOS Format  M-A Append     M-B Backup File ^T Browse
^C Cancel    M-M Mac Format  M-P Prepend    M-Q Discard buffer

```

OUTPUT:

```
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp29$ nano CMakeLists.txt
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp29$ ls
CMakeLists.txt  priority_interrupt.cpp
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp29$ mkdir build
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp29$ cd build
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp29/build$ cmake ..
-- The C compiler identification is GNU 15.2.0
-- The CXX compiler identification is GNU 15.2.0
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /usr/bin/cc - skipped
-- Detecting C compile features
-- Detecting C compile features - done
-- Detecting CXX compiler ABI info
-- Detecting CXX compiler ABI info - done
-- Check for working CXX compiler: /usr/bin/c++ - skipped
-- Detecting CXX compile features
-- Detecting CXX compile features - done
-- Configuring done (5.1s)
-- Generating done (0.0s)
-- Build files have been written to: /home/sai_varshini/systemc_test/exp29/build
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp29/build$ make
[ 50%] Building CXX object CMakeFiles/priority_interrupt.dir/priority_interrupt.cpp.o
[100%] Linking CXX executable priority_interrupt
[100%] Built target priority_interrupt
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp29/build$ ./priority_interrupt

SystemC 3.0.2-Accellera --- Nov  5 2025 12:44:46
Copyright (c) 1996-2025 by all Contributors,
ALL RIGHTS RESERVED
Low Priority Interrupt at 1 s
High Priority Interrupt at 2 s
sai_varshini@DESKTOP-5G7BJMJ:~/systemc_test/exp29/build$ |
```